A. k-Factorization

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1100 | Tags: implementation, math, number theory | https://codeforces.com/problemset/problem/797/A

Given a positive integer n, find k integers (not necessary distinct) such that all these integers are strictly greater than 1, and their product is equal to n.

Input

The first line contains two integers n and k ($2 \le n \le 100000$, $1 \le k \le 20$).

Output

If it's impossible to find the representation of n as a product of k numbers, print -1.

Otherwise, print k integers in any order. Their product must be equal to n. If there are multiple answers, print any of them.

Examples

Input
100000 2

Output
2 50000

Input
100000 20

Output
-1

Input
1024 5

Output
2 64 2 2 2